

ASPIRE

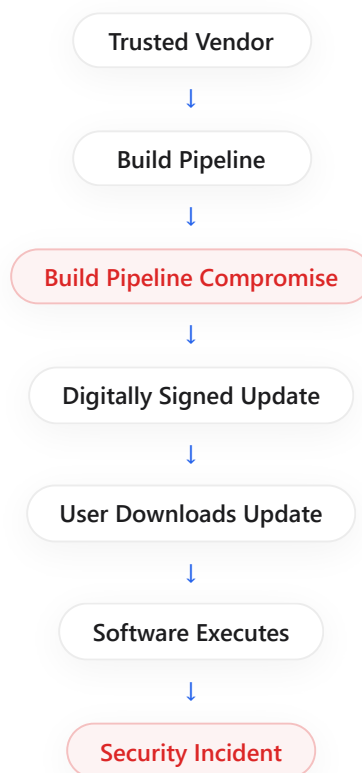
Artifact Security and Portable Executable Intelligence for Risk Evaluation

"Pre-execution software artifact trust evaluation using Portable Executable intelligence and security indicator correlation."

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Research Motivation

Supply-chain attacks such as **SolarWinds** and **3CX** show that trusted software updates can become attack vectors. Existing defenses usually detect threats *after* execution. ASPIRE focuses on evaluating trust **before** the file runs.



Research Question

Can software artifacts be evaluated and assigned an explainable trust score before execution, even when they originate from trusted vendors?

This forms the central research problem investigated by ASPIRE.

Traditional Detection:
After Execution ❌

ASPIRE:
Before Execution ✅

Why ASPIRE



Artifact Security

Security assessment of software artifacts such as **.EXE**, **.DLL**, and **.SYS** files.



Portable Executable Intelligence

Extraction of security-relevant features from Windows PE files.



Risk Evaluation

Generation of an explainable trust score for allow / review / block decisions.

High-Level Architecture

1

Software Artifact

Input executable, library, or driver submitted for analysis.

2

Artifact Fingerprinting

Creates a unique artifact profile.

3

PE Structural Analysis

Extracts executable characteristics.

4

SBOM & Dependency Analysis

Evaluates software components and dependencies.

5

Risk Indicator Generation

Generates security-relevant indicators.

6

Pre-Execution Correlation

Correlates indicators before execution.

7

ML-based Anomaly Assessment

Identifies anomalous artifacts.

8

RISK FUSION ENGINE

Computes an explainable trust score.

9

Risk Classification

Produces the final risk decision.

SBOM = Software Bill of Materials

Proposed Contribution

Artifact Fingerprinting for Software Trust Analysis

Portable Executable Intelligence for Security Assessment

Pre-Execution Security Indicator Correlation

Risk Fusion for Explainable Trust Evaluation

Research Inspiration - Base Paper

PDF

EMBER: An Open Dataset for Training Static PE Malware Machine Learning Models
Hyrum S. Anderson & Phil Roth

This paper introduces a large benchmark dataset for Portable Executable malware analysis and demonstrates machine-learning-based detection using PE features. It forms the research foundation for the Portable Executable Intelligence component of ASPIRE.

Expected Outcome

- Generate artifact fingerprints
- Extract Portable Executable intelligence
- Analyze software dependencies through SBOM
- Correlate security indicators before execution
- Generate an explainable trust score

ASPIRE aims to shift software trust evaluation earlier in the lifecycle through pre-execution analysis and risk-based assessment.

Conclusion

ASPIRE investigates a pre-execution approach for software trust evaluation by combining Portable Executable intelligence, dependency analysis, security indicator correlation, and risk-based assessment.